

Guide

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A. Setup

Here are the steps on getting the BC up and running on your local OpenFoam-12. Open a terminal and do the following:

1. Clone the repository to any folder (given that you have access) by typing:
`git clone git@gitlab.sintef.no:Hursanay.Fyhn/syntheticTurbulenceinlet.git`
2. Go into the cloned folder by typing
`cd syntheticTurbulenceInlet`
3. Source you local OpenFoam-12 the way you normally source it.
4. Compile the boundary condition and add it to the list of boundary conditions for your OpenFoam-12 by typing:
`wmake`
(ignore the message about .H files already existing)
5. You can now use the BC like any other, just remember to source the relevant library by adding this line to your controlDict:
`libs ("libsyntheticTurbulenceInlet.so");`
6. Within the cloned directory, you will find a folder titled tutorials. Try out the tutorial cases for the four different geometry modes inside the tutorials folder.

B. Parameters

- *geometryMode* (required)
Choose between annulus, disk, rectangle or channel
- *wallLongOrShort* (optional)
When *geometryMode* is channel, choose if the wall is on the long or the short side.
Default: "long".
- *Umean* (required)
Mean speed in the main flow direction and can be a scalar or a table.
- *Uscale* (optional)
This corresponds to the mean speed and is used to initialize some properties for the eddies. If your *Umean* is a table and its first element is far from the mean of your list, then you should specify the mean here.
Default: The first element in *Umean*
- *Utau* (optional)
Friction velocity
Default: *Utau* is calculated from the expression for a smooth pipe. There is an absolute lower limit that maps the entire DNS profile over the entire width, and it can be forced by e.g. choosing a negative value.
- *swirlNr* (optional)
Swirl number
Default: 0
- *d* (optional)
Eddy density (ratio of the sum of eddy volumes to the eddy box volume) Default: 1
- *scale* (optional)
A scaling factor for U'
Default: 1

- *nCellPerEddy* (optional)
Minimum eddy length in units of number of cells
Default: 5
- *Lref* (optional)
Normalisation factor for integral-length scale
Default: 1
- *seed* (optional)
Seed for the random numbers
Default: an arbitrary number
- *value* (required)
Initial value for velocity field and is overwritten during the setup stage.

C. Example usage

```
inlet
{
type syntheticTurbulenceInlet;
value (10 0 0);
geometryMode annulus;
Umean 10;
}
```